Original MCQs

Question 362 (Graph Theory – nan | LLM difficulty:)

The complete bipartite graph $K_{m,n}$ has:

A. $m \times n$ edges.

B. $m + n$ edges.

C. $oddcycles$.

D. $m \times n$ vertices.

Question 365 (Graph Theory – nan | LLM difficulty:)

The graph in the picture

A. $isHamiltonian$.

B. $isbipartite$.

C. $iscomplete$.

D. $isEulerian$.

Question 370 (Graph Theory – nan | LLM difficulty:)

Which of the following graphs admits an Euler circuit?

A. G_1 and G_3.

B. $Allgraphs$.

C. $None of the graphs$.

D. G_1 and G_2.

Question 372 (Graph Theory – nan | LLM difficulty:)

Which of the following sentences is false?

A. $Thegraphhasacircuit$.

B. $Thegraphhasacycle$.

C. $ThegraphisEulerian$.

D. $ThegraphisHamiltonian$.

Question 373 (Graph Theory – nan | LLM difficulty:)

The complete graph K_n is:

A. Eulerian for odd values of n.

B. Hamiltonian for all values of n.

C. Eulerian for even n.

D. bipartite for all n.

Question 374 (Graph Theory – nan | LLM difficulty:)

In a tree with six vertices

A. $therearefiveedges.$

B. $allverticeshavedegreethree.$

C. $therearetwoleafs, thatis, twoverticeswithdegreeone.$

D. $thereisacycle.$

Question 386 (Graph Theory – nan | LLM difficulty:)

How many non-isomorphic spanning trees are there in the graph?

A. 3

B. 4

C. 5

D. 6

Question 465 (Graph Theory – nan | LLM difficulty:)

A simple graph is a graph, oriented or not, with

A. $anonemptysetofvertices.$

B. $multipleedges.$

C. $loops.$

D. $anonemptysetofedges.$

Question 466 (Graph Theory – nan | LLM difficulty:)

The graph

A. $hasatleasttwonon-adjacentvertices.$

B. $isdisconnected.$

C. has one vertex with degree 5.

D. $hasanisolatedvertex.$

Question 468 (Graph Theory – nan | LLM difficulty:)

If a subgraph $G'(V', E')$ of the graph $G(V, E)$ is regular, then:

A. $V' \subseteq V$, $E' \subseteq E$ and all vertices of G' have the same degree.

B. $V' = V$ and $E' \subseteq E$.

C. $V \subseteq V'$ and $E \subseteq E'$.

D. $V' \subseteq V$ and $E' = \emptyset$.

Question 469 (Graph Theory – nan | LLM difficulty:)

If a simple graph has ten edges and five vertices with same degree, then each vertex has degree

A. $four.$

B. $two.$

C. $three.$

D. $five.$

Question 470 (Graph Theory – nan | LLM difficulty:)

If a simple graph has at least one edge, then

A. $therearealwaystwoverticeswithsamedegree.$

B. $therearealwaystwoverticeswithodddegree.$

C. $therearenoverticeswithsamedegree.$

D. $therealwaysaretwoverticeswithevendegree.$

Question 471 (Graph Theory – nan | LLM difficulty:)

G is a non trivial connected simple graph. Which is the true sentence?

A. If $|V(G)| \geq 2$ there is a connected induced subgraph.

B. All induced subgraphs of G are connected.

C. All subgraphs of G are connected.

D. $There is a non-trivial proper induced subgraph that is also connected.$

Question 472 (Graph Theory – nan | LLM difficulty:)

If $G = (V, E)$ is a k-regular graph, then

A. $2|E| = k|V|$.

B. $|E| = k|V|$.

C. $|E| = 2k|V|$.

D. $|E| = 2|V|$.

Question 473 (Graph Theory – nan | LLM difficulty:)

Which pair has two isomorphic graphs?

A. (a).

B. (b).

C. (c).

D. (d).

Question 474 (Graph Theory – nan | LLM difficulty:)

How many non-isomorphic simple graphs with four vertices are there?

A. 12

B. 10

C. 16

D. 20

Question 475 (Graph Theory – nan | LLM difficulty:)

The adjacency matrix $\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$ represents the graph

A. (b).

B. (a).

C. (c).

D. (d).

Question 476 (Graph Theory – nan | LLM difficulty:)

The adjacency matrix of the attached directed graph is

A. $\begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{pmatrix}$

B. $\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}$

C. $\left(\begin{matrix} 0 & 1 & 0 & 0 & & 1 \\ 1 & 1 & 0 & 1 & & 1 \\ 0 & 0 & 0 & 0 & & 1 \\ 0 & 1 & 0 & 0 & & 0 \\ 1 & 1 & 1 & 0 & 0 endpmatrix & \\ \textbf{C.} \begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 1 \\ 1 & -1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{pmatrix} \end{matrix}\right.$

Question 477 (Graph Theory – nan | LLM difficulty:)

An orientation of a graph can be achieved by replacing each non ordered pair of vertices $\{x, y\}$, which is an edge, by one of the ordered pairs (x, y) or (y, x).How many non isomorphic orientations of K_3 are there?

A. 2

B. 1

C. 3

D. 4

Question 478 (Graph Theory – nan | LLM difficulty:)

The attached graph has incidence matrix:

A. $\begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$

B. $\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$

C. $\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{pmatrix}$

D. $\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}$

Question 479 (Graph Theory – nan | LLM difficulty:)

A bridge uv is an edge of a connected graph whose removal leaves the graph disconnected. Which is the true sentence?

A. $Everyedgeinatreeisabridge.$

B. No circuit includes the bridge uv.

C. Vertex u and vertex v do not have adjacent vertices in common.

D. Vertex u and vertex v have the same degree.

Question 480 (Graph Theory – nan | LLM difficulty:)

The complete graph $K_{m,n}$ has:

A. $m \times n$ edges.

B. $m + n$ edges.

C. $oddcycles.$

D. $m \times n$ vertices.

Question 481 (Graph Theory – nan | LLM difficulty:)

Which of the following sentences is not equivalent to the other three for a connected graph G?

A. G contains a tree.

B. G is Eulerian.

C. $Thedegreeofeveryvertexiseven.$

D. G is a disjoint union of cycles.

Question 482 (Graph Theory – nan | LLM difficulty:)

How many more edges are there in the complete graph K_7 than in the completegraph K_5?

A. 11

B. 10

C. 12

D. 9

Question 581 (Graph Theory – nan | LLM difficulty:)

Knowing that T is a tree, which of the following sentences is false?

A. T has m edges and $m - 1$ vertices.

B. T is connected and contains no cycles.

C. There is precisely one path between any two vertices of T.

D. T is a bipartite graph if it has at least two vertices.

Question 582 (Graph Theory – nan | LLM difficulty:)

Which of the following is an example of a Hamiltonian family of graphs?

A. K_n with $n \geq 3$.

B. $K_{m,n}$ with $m,\ n \geq 1$.

C. $All3 - regulargraphs.$

D. $Allbipartitegraphs.$

Question 583 (Graph Theory – nan | LLM difficulty:)

The length of a walk is the number of edges in it. A graph G is bipartite if and only if:

A. There are two disjoint non-empty vertex subsets, V_1 and V_2 such that each edge joins a vertex in V_1 to a vertex in V_2.

B. $Therearenocyclesofevenlength.$

C. G admits a partition on the vertex set.

D. $Therearenopathsofoddlength.$

Question 584 (Graph Theory – nan | LLM difficulty:)

The incidence matrix $\begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$ represents the graph:

A. $d.$

B. $c.$

C. $a.$

D. $b.$

Question 589 (Graph Theory – nan | LLM difficulty:)

If G is a graph with n vertices and m edges without loops nor multiple edges, how manyedges does the complement of G have?

A. $\frac{n(n-1)}{2} - m$ edges.

B. $\frac{m(m-1)}{2} - n$ edges.

C. 24 edges.

D. $n(n-1) - m$ edges.

Question 924 (Graph Theory – nan | LLM difficulty:)

Consider a connected planar graph with 8 vertices, each of degree 3. Into how many regions (faces) is the plane divided by a planar representation of such graph?

A. 6

B. 3

C. 8

D. 1

Question 929 (Graph Theory – nan | LLM difficulty:)

The number of vertices, $|V|$, edges, $|E|$, and faces, $|F|$, of the attached graph is

A. $|V| = 5$, $|E| = 7$ and $|F| = 4$.

B. $|V| = 5$, $|E| = 6$ and $|F| = 7$.

C. $|V| = 5$, $|E| = 5$ and $|F| = 5$.

D. $|V| = 5$, $|E| = 5$ and $|F| = 3$.

Question 930 (Graph Theory – nan | LLM difficulty:)

Which of the attached graphs are planar?

A. G_1, G_2 and G_3.

B. $Allgraphs.$

C. G_1 and G_3.

D. G_2 and G_4.

Question 933 (Graph Theory – nan | LLM difficulty:)

For which m and n the graph $K_{m,n}$ is not planar?

A. $n \geq 3$ and $m \geq 3$.

B. $n = m$.

C. $n \geq 3$.

D. $m \geq 3$.

Question 934 (Graph Theory – nan | LLM difficulty:)

Consider the graphs in the image. About the chromatic number, we can say that:

A. $\chi(G_1) = \chi(G_3) = 3$ and $\chi(G_2) = \chi(G_4) = 4$.

B. $\chi(G_1) = \chi(G_3) = 4$ and $\chi(G_2) = \chi(G_4) = 3$.

C. All graphs have chromatic number 3.

D. All graphs have chromatic number 4.

Question 935 (Graph Theory – nan | LLM difficulty:)

The chromatic number of the graph K_n is:

A. n.

B. $n - 1$.

C. $\frac{n(n-1)}{2}$.

D. 2.

Question 936 (Graph Theory – nan | LLM difficulty:)

The chromatic number of the graph $K_{m,n}$ is:

A. 2.

B. $n \times m$.

C. $n + m$.

D. m.

Question 937 (Graph Theory – nan | LLM difficulty:)

Consider the following statements about graph G:

- **A.** The statement $i.$ is true and the statement $ii.$ is false.
- **B.** The statement $i.$ is false and the statement $ii.$ is true.
- **C.** $The statements are both true.$
- **D.** $The statements are both false.$

Question 938 (Graph Theory – nan | LLM difficulty:)

Which of the following sentences is true?

- **A.** A planar graph with chromatic number 2 must be Hamiltonian.
- **B.** A Hamiltonian graph with chromatic number 2 must be planar.
- **C.** A graph which is both Hamiltonian and Eulerian and has chromatic number 2 must be planar.
- **D.** A graph which is both Hamiltonian and Eulerian must have chromatic number less than 6.

Question 939 (Graph Theory – nan | LLM difficulty:)

The graph T is a tree. Consider the following statements:

- **A.** $All statements are true.$
- **B.** Only statements $i.$ and $ii.$ are true.
- **C.** Only statements $ii.$ and $iii.$ are true.
- **D.** $All statements are false.$

Question 940 (Graph Theory – nan | LLM difficulty:)

Consider the following weighted graph. We can say that:

- **A.** The path af is shorter than ag.
- **B.** The path cf is shorter than bg.
- **C.** The path cz is shorter than bz.
- **D.** The path dg is shorter than ef.

Question 941 (Graph Theory – nan | LLM difficulty:)

Consider the following weighted graph and indicate the shortest path between a and z.

- **A.** $acdgz$.
- **B.** $acegz$.
- **C.** $abdfz$.
- **D.** $abdegz$.

Question 942 (Graph Theory – nan | LLM difficulty:)

How many spanning trees has the following graph?

- **A.** 8.
- **B.** 7.
- **C.** 2.
- **D.** 1.

Question 943 (Graph Theory – nan | LLM difficulty:)

A spanning tree of the the following connected graph is the set of edges and terminal vertices:

- **A.** v_1v_5, v_2v_5, v_3v_5, v_4v_5 and v_6v_5.
- **B.** v_1v_2, v_2v_3, v_3v_4, v_4v_5, v_5v_6 and v_6v_1.
- **C.** v_1v_2, v_1v_5, v_1v_6, v_4v_3 and v_4v_5.
- **D.** v_3v_2, v_3v_5, v_3v_4 and v_4v_5.

Question 944 (Graph Theory – nan | LLM difficulty:)

Consider the following weighted graph. Applying the Kruskal algorithm to find the minimum spanning tree starting at vertex a, the order in which edges are added to form a tree is:

- **A.** $\{d,\ f\}$, $\{a,\ c\}$, $\{a,\ b\}$, $\{c,\ d\}$, $\{d,\ e\}$.
- **B.** $\{a,\ c\}$, $\{a,\ b\}$, $\{c,\ d\}$, $\{d,\ f\}$, $\{d,\ e\}$.
- **C.** $\{a,\ b\}$, $\{b,\ c\}$, $\{c,\ d\}$, $\{d,\ e\}$, $\{e,\ f\}$.
- **D.** $\{a,\ c\}$, $\{b,\ c\}$, $\{a,\ c\}$, $\{c,\ e\}$, $\{d,\ f\}$.

Question 945 (Graph Theory – nan | LLM difficulty:)

Consider the following weighted graph. Applying the Prim algorithm to find the minimum spanning tree starting at vertex a, the order in which edges are added to form a tree is:

A. $\{a,\ c\}$, $\{a,\ b\}$, $\{c,\ d\}$, $\{d,\ f\}$, $\{d,\ e\}$.

B. $\{a,\ b\}$, $\{b,\ c\}$, $\{c,\ d\}$, $\{d,\ e\}$, $\{e,\ f\}$.

C. $\{a,\ c\}$, $\{b,\ c\}$, $\{a,\ c\}$, $\{c,\ e\}$, $\{d,\ f\}$.

D. $\{d,\ f\}$, $\{a,\ c\}$, $\{a,\ b\}$, $\{c,\ d\}$, $\{d,\ e\}$.

Question 946 (Graph Theory – nan | LLM difficulty:)

The Dijkstra algorithm determines

A. $the shortest path between two given vertices in a weighted connected graph.$

B. $the minimum spanning tree.$

C. $a spanning tree.$

D. $all paths between two given vertices.$

Question 947 (Graph Theory – nan | LLM difficulty:)

The table shows the distances between 5 European cities.Considering the graph K_5 where the vertices are those 5 European cities, the corresponding minimum spanning tree is:

A. $\{LT,\ RO\}$, $\{PT,\ IT\}$,$\{IT,\ LT\}$, $\{PT,\ IE\}$.

B. $\{LT,\ RO\}$, $\{PT,\ IT\}$,$\{IT,\ LT\}$.

C. $\{PT,\ IT\}$, $\{IT,\ RO\}$, $\{RO,\ LT\}$, $\{LT,\ IE\}$.

D. $\{IT,\ RO\}$, $\{PT,\ IT\}$, $\{PT,\ IE\}$ $\{PT,\ RO\}$.

Question 1023 (Graph Theory – nan | LLM difficulty:)

In a simple graph of order n the maximum possible degree of a vertex is:

A. $n - 1$.

B. n.

C. $2n$.

D. $\frac{n(n-1)}{2}$.

Question 1501 (Graph Theory – nan | LLM difficulty:)

Among a group of 5 people, is it possible for everyone to be friends with exactly 2 of the people in the group?What about 3 of the people in the group?

- **A.** It is possible for 2, but not for 3.
- **B.** It is possible for 3, but not for 2.
- **C.** It is possible for both 2 and 3.
- **D.** It is not possible for both 2 and 3.

Question 1504 (Graph Theory – nan | LLM difficulty:)

Let $G = (V, E)$ be a simple graph such that $V = [10]$ and E is defined by

$$\{a, b\} \in E \iff \text{a} \neq b \text{ and } a - b \text{ is a multiple of } 2$$

Consider the following sentences: G is connected.- G is bipartite.- $|E|$ is a multiple of 4.- G is Hamiltonian.: $[10] = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$Choose the correct option:

- **A.** $Onlyoneofthesentencesiscorrect.$
- **B.** $Thereareexactlytwocorrectsentences.$
- **C.** $Thereareexactlythreecorrectsentences.$
- **D.** $Allthesentencesaretrue.$

Question 1505 (Graph Theory – nan | LLM difficulty:)

Is it true that all trees represent a bipartite graph?

- **A.** $Yes.$
- **B.** $No.$
- **C.** $Itdependsonthetree.$
- **D.** $Toanswerthisquestion, thetreedrawingisrequired.$

Question 1507 (Graph Theory – nan | LLM difficulty:)

Which of the following graphs is not connected, knowing that all of them have 5 vertices?

$$\phi_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ \{1,2\} & \{1,2\} & \{2,3\} & \{3,4\} & \{1,5\} & \{1,5\} \end{pmatrix}$$

$$\phi_2 = \begin{pmatrix} b & a & e & d & c & f \\ \{4,5\} & \{1,3\} & \{1,3\} & \{2,3\} & \{2,5\} & \{4,5\} \end{pmatrix}$$

$$\phi_3 = \begin{pmatrix} b & f & e & d & c & a \\ \{4,5\} & \{1,3\} & \{1,3\} & \{2,3\} & \{2,4\} & \{4,5\} \end{pmatrix}$$

$$\phi_4 = \begin{pmatrix} a & b & c & d & e & f \\ \{1,2\} & \{2,3\} & \{1,2\} & \{2,3\} & \{3,4\} & \{1,5\} \end{pmatrix}$$

$$\phi_5 = \begin{pmatrix} a & b & c & d & e & f \\ \{1,2\} & \{2,3\} & \{1,2\} & \{1,3\} & \{2,3\} & \{4,5\} \end{pmatrix}$$

A. Graph ϕ_5.

B. Graphs ϕ_2 and ϕ_3.

C. Graph ϕ_1.

D. $All of them are connected.$

Question 1508 (Graph Theory – nan | LLM difficulty:)

In which of the following graphs can we find an Eulerian circuit (a trail on G that includes every edge of G such that the only repeated vertices are the start and the end)?

$$\phi_1 = \begin{pmatrix} F & B & C & D & E & A \\ \{1,2\} & \{1,2\} & \{2,3\} & \{3,4\} & \{4,5\} & \{4,5\} \end{pmatrix}$$

$$\phi_2 = \begin{pmatrix} b & f & e & d & c & a \\ \{4,5\} & \{1,3\} & \{1,3\} & \{2,3\} & \{2,4\} & \{4,5\} \end{pmatrix}$$

$$\phi_3 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ \{1,2\} & \{1,2\} & \{2,3\} & \{3,4\} & \{4,5\} & \{4,5\} \end{pmatrix}$$

$$\phi_4 = \begin{pmatrix} b & a & e & d & c & f \\ \{4,5\} & \{1,3\} & \{1,3\} & \{2,3\} & \{2,5\} & \{4,5\} \end{pmatrix}$$

$$\phi_5 = \begin{pmatrix} a & b & c & d & e & f \\ \{1,3\} & \{3,4\} & \{1,2\} & \{2,3\} & \{3,5\} & \{4,5\} \end{pmatrix}$$

A. Graph ϕ_5.

B. Graphs ϕ_1 and ϕ_2.

C. $None of them has an Eulerian circuit.$

D. $All of them have an Eulerian circuit.$

Question 1509 (Graph Theory – nan | LLM difficulty:)

How many simple graphs with n vertices and e edges are there?

- **A.** $\binom{\binom{n}{2}}{e}$
- **B.** $2^{\binom{n}{e}}$
- **C.** $\binom{n}{e}$
- **D.** 2^e

Question 1510 (Graph Theory – nan | LLM difficulty:)

How many simple graphs with n vertices are there?

- **A.** $2^{\binom{n}{2}}$.
- **B.** 2^n.
- **C.** $2n$.
- **D.** $\binom{n}{2}$.

Question 1511 (Graph Theory – nan | LLM difficulty:)

Consider the graph G whose adjacency matrix is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

How many connected component does G have?

- **A.** 2.
- **B.** 1.
- **C.** 3.
- **D.** 4.

Question 1512 (Graph Theory – nan | LLM difficulty:)

Let G be a connected graph and $e \in E(G)$, where $E(G)$ denotes the set of edges of G.Assume that e is in a cycle of G.What can we say about $G - e$ (the graph G without the edge e)?

A. $G - e$ is also connected.

B. $G - e$ is not connected.

C. $We don't have enough information to conclude anything.$

D. $To answer this question, a graph drawing is required.$

Question 1643 (Graph Theory – nan | LLM difficulty:)

The adjacency matrix that represents the graph

A. $\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$

B. $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$

D. $\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$

Question 1644 (Graph Theory – nan | LLM difficulty:)

The graph

A. $is disconnected.$

B. $has one vertex with degree three.$

C. $has every vertices are adjacent.$

D. $has every vertices are adjacent.$

Question 1646 (Graph Theory – nan | LLM difficulty:)

The incidence matrix $\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \end{bmatrix}$ represents the graph:

A. (c)

B. (d)

C. (a)

D. (b)

Question 1647 (Graph Theory – nan | LLM difficulty:)

Which pair of simple graphs are isomorphic?

A. (c)

B. (b)

C. (d)

D. (a)

Question 1648 (Graph Theory – nan | LLM difficulty:)

Consider the following weighted graph and, only one do not indicate the shortest path between v_0 and v_3 using Dijkstra's algorithm.

A. $v_0v_4v_8v_3$

B. $v_0v_1v_2v_3$

C. $v_0v_4v_2v_3$

D. $v_0v_4v_1v_2v_3$

Question 1649 (Graph Theory – nan | LLM difficulty:)

Consider the weighted graph in the picture. Applying the Prim's algorithm to find the minimum spanning tree starting at vertex a, the order in which edges are added to form a tree is:

A. $\{a,\ d\}$, $\{d,\ e\}$, $\{d,\ c\}$, $\{d,\ b\}$

B. $\{e,\ d\}$, $\{d,\ c\}$, $\{b,\ d\}$, $\{a,\ d\}$

C. $\{a,\ b\}$, $\{b,\ e\}$, $\{e,\ d\}$, $\{d,\ c\}$

D. $\{c,\ d\}$, $\{d,\ e\}$, $\{e,\ b\}$, $\{b,\ d\}$

Question 1650 (Graph Theory – nan | LLM difficulty:)

How many non isomorphic spanning trees has the following graph?

A. 2

B. 7

C. 8

D. 1

Question 1651 (Graph Theory – nan | LLM difficulty:)

Which of the following graphs is planar?

A. G_1

B. G_2

C. *none*

D. *both*

Question 1652 (Graph Theory – nan | LLM difficulty:)

The indidence matrix of the following graph is

A. $\begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 1 \\ 1 & -1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{bmatrix}$

B. $\begin{bmatrix} 0 & -1 & 0 & 0 & 1 \\ 1 & 1 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}$

D. $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$

Question 1872 (Graph Theory – nan | LLM difficulty:)

If the graph G has chromatic number $\chi(G) = 2$, then

A. $thegraphisbipartite.$

B. $thegraphisplanar.$

C. $thegraphiscomplete.$

D. $thegraphisconnected.$

Question 1875 (Graph Theory – nan | LLM difficulty:)

How many non isomorphic trees with five vertices exists?

A. 5

B. 8

C. 1

D. 9

Question 2148 (Graph Theory – nan | LLM difficulty:)

A simple graph with eight vertices of degree 3 has:

- **A.** $12 edges.$
- **B.** $11 edges.$
- **C.** $24 edges.$
- **D.** $6 edges.$

Question 2150 (Graph Theory – nan | LLM difficulty:)

A simple graph where all vertices have the same degree is a:

- **A.** $regular graph.$
- **B.** $complete graph.$
- **C.** $bipartite graph.$
- **D.** $disconnected graph.$

Question 2158 (Graph Theory – nan | LLM difficulty:)

The length of a walk is the number of edges in it. A graph G is bipartite if and only if:

- **A.** $there is no cycles of odd length.$
- **B.** $there is no cycles of even length.$
- **C.** G admits a partition on vertex set.
- **D.** Exists two disjoint non empty vertex subsets, V_1 and V_2, that each edge joins a vertex in V_1 to a vertex in V_2.